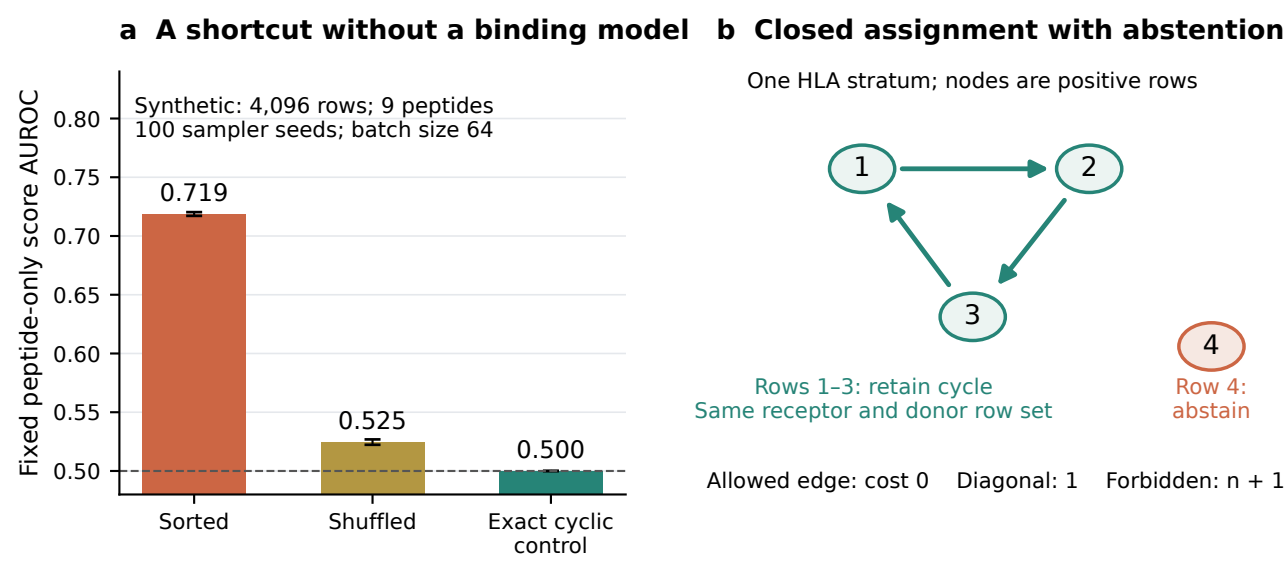
a A shortcut without a binding model
Synthetic: 4,096 rows; 9 peptides
100 sampler seeds; batch size 64
Fixed peptide-only score AUROC
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.719
0.525
0.500
Sorted
Shuffled
Exact cyclic control
b Closed assignment with abstention
One HLA stratum; nodes are positive rows
1
2
3
4
Rows 1–3: retain cycle
Same receptor and donor row set
Row 4: abstain
Allowed edge: cost 0
Diagonal: 1
Forbidden: n + 1